

CONFIDENTIAL — ARCHITECTURE

Quenyx Platform Architecture Bible

Quenyx vOPS HUB — v1.0.0 RC1 · Document v2.1

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REVISION HISTORY

Version	Date	Notes
1.0	2026	Initial architecture bible (through Sprint 19).
2.0	2026-06-29	RC1 alignment: native QynSight engines, QynCore internal communication model, Integrations = external only, shared platform AI.
2.1	2026-06-30	Sprint 21 — QynSight Operations Intelligence: QynSight becomes a live AI consumer reusing the shared runtime; UUID-only <code>/api/qynsight/intelligence/*`</code> surface.

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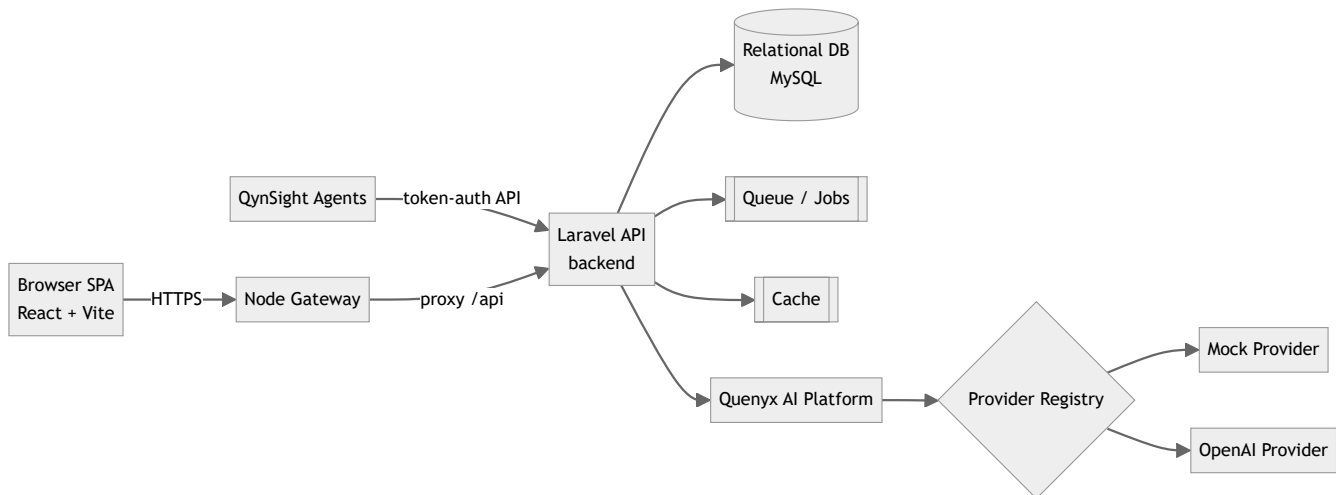
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Audience: Architects, senior engineers, auditors. **Status basis:** v1.0.0 RC1. Diagrams reflect the **current** production code.

1. Platform architecture (high level)

Quenyx vOPS HUB is a **monorepo** with three runtimes — a Laravel API backend, a React SPA frontend, and a Node gateway — over a relational database, queue, and cache.



2. Monorepo structure

Path	Runtime	Responsibility
<code>backend/</code>	Laravel (PHP 8.3)	API, services, QCIF engines, AI platform, persistence
<code>frontend/</code>	React + Vite + TS	SPA UI, module registry, workspace UX
<code>gateway/</code>	Node	Edge/proxy layer in front of the API
<code>agent/</code>	—	QynSight host agent artifacts
<code>docs/</code>	—	Documentation (this pack lives in <code>docs/quenyx-v1/</code>)
<code>scripts/</code>	—	Operational scripts

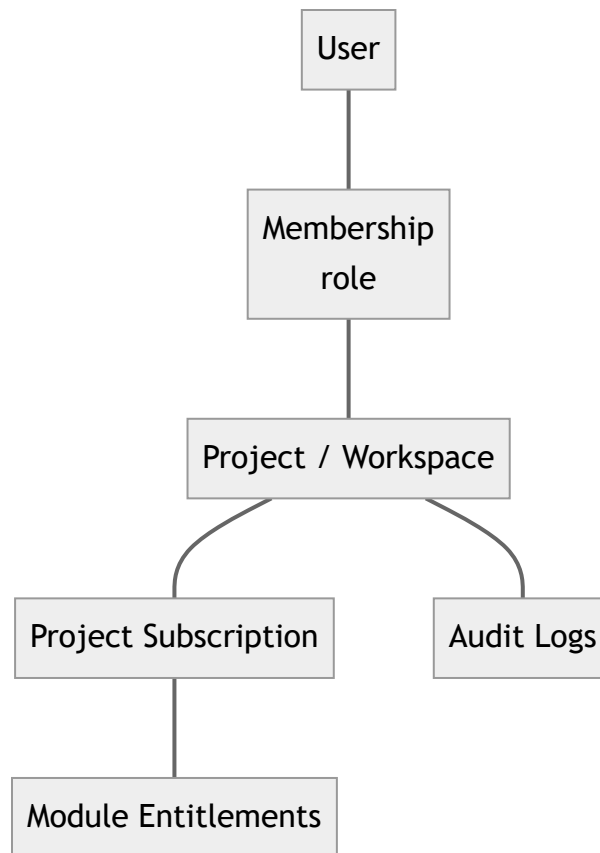
3. Backend / frontend / gateway

- **Backend** — `routes/api.php` is the single entry, requiring per-domain route files (`compliance-corpus.php` , `compliance-graph.php` , `ai-orchestration.php` , `compliance-copilot.php` , `compliance-rag.php` , `compliance-executive.php` , `quenyx-ai.php` , ...). Controllers are thin; business logic lives in `app/Services/**` .

- **Frontend** — React SPA; module/route registry in `src/constants/platformRegistry.ts`; layout and sidebar in `src/layouts/AppLayout.tsx`.
- **Gateway** — Node service in front of the API (edge concerns / proxy). Build & run on the server (see Doc 10); not buildable in the audit sandbox (no Node).

4. Workspace / project model

The tenant boundary is the **project** (a.k.a. **workspace** — the two are API aliases pointing to the same controllers, see `routes/api.php`). Users belong to projects via **memberships** with roles; modules are entitled per project via **subscriptions** and **module overrides**.



5. RBAC and entitlements

- **Authentication:** Laravel **Sanctum**; all tenant routes live under `auth:sanctum`.
- **Authorization:** project membership enforced via `ProjectPolicy` (controllers call `authorize('view', $project)`).
- **Module entitlements:** middleware gates feature areas:
 - `project.module:qynsight` → QynSight `observe/*` routes.
 - `project.qynshield` → all QynShield/QCIF routes (corpus workspace, copilot, evidence, gap, recommendations, retrieval, RAG, executive).
- **Overrides:** `ProjectModuleOverrideController` allows per-project module access overrides (audited).

6. Module registry

The **frontend** registry (`platformRegistry.ts`) defines all module entries. The **business modules** are: QynSight, QynShield, QynAsset, QynRun, QynKnow, QynNotify, QynReact, QynVA, QynSupport, and QynBalance. Sidebar **visibility** is a **separate** concern controlled by `HIDE_NON_QYNSIGHT_MODULES = true` + `ACTIVE_MODULE_KEYS = ['qynsight']` via `isModuleTemporarilyVisible()` . Hidden modules are **registered platform modules disabled in the navigation by a sidebar flag** — not removed; flipping the flag restores them. The **backend** additionally tracks module AI-readiness in `config/quenyx_ai.php` + `QuenyxModuleCatalog` (production / reserved / planned), independent of UI.

QynCore and Integrations are not business modules. QynCore is the **platform core** — the internal services layer through which modules communicate (see §6.1). Integrations is a **platform page** for **external** systems only (see §6.2). As of **RC1.1**, the frontend business catalog (`platformRegistry.ts`) and the AI module universe (`config/quenyx_ai.php`) no longer list `qynintegrations` as a business module, and `qyncore` is documented there as the platform core. The `qyncore` / `qynintegrations` keys are **retained only as entitlement keys** (plans, subscriptions, and the gateway gate for `/integrations*`) for backward compatibility — never presented as business modules.

6.1 Internal communication — QynCore platform services

Modules **never** talk to each other over HTTP, webhooks, or the Integrations page. Internal communication is **platform-native**, brokered by **QynCore**:

QynCore service	Responsibility	Backed today by
Platform Event Bus	Asynchronous in-platform events between modules	Laravel events + <code>jobs</code> queue
Domain Events	Module-level domain notifications	Eloquent/domain events
Shared Services	Cross-module services (identity, billing/subscription, common AI)	<code>app/Services/**</code>
Module Registry	Known modules + entitlement state	<code>platformRegistry.ts</code> , subscriptions/overrides
Service Registry	Discoverable platform services/capabilities	<code>QuenyxAiPlatform</code> capability catalog
AI Context Broker	Workspace-scoped context handed to the AI platform	<code>AiContext</code> / <code>QuenyxAiPlatform</code>
Permission Broker	Centralized authorization decisions	<code>ProjectPolicy</code> + entitlement middleware
Audit Pipeline	Structured, immutable audit of sensitive actions	<code>audit_logs</code> + audit loggers
Notification Broker	In-platform notification fan-out	notification/event surfaces
Workspace Context	Tenant (project/workspace) resolution + isolation	project↔workspace resolver

Modules communicate **automatically through the platform**; there is no module-to-module HTTP, webhook, or integration dependency.

6.2 Integrations (platform page, external only)

The **Integrations** page connects Quenyx to **external** systems — e.g. Microsoft, Azure, AWS, OCI, Google Cloud, GitHub, GitLab, ServiceNow, Jira, Fortinet, Cisco, VMware, Slack, Teams, Splunk, Elastic, Wazuh, REST APIs, Webhooks, LDAP, Active Directory, Entra ID, SMTP. It is **not** a module and carries **no** internal module-to-module traffic. (Legacy Nagios, where still in use, is one such optional external integration / migration source — never a platform dependency.)

7. QynSight architecture (native monitoring)

QynSight ("observe") provides **native** monitoring — there is **no external monitoring daemon and no Nagios dependency**. It is composed of native engines:

- **Discovery Engine** — target/host discovery and port scans.
- **Monitoring Engine** — runs service checks (HTTP, TCP, Ping, and custom plugin scripts) in-platform via the scheduled `observe:run-checks` command (`engine_key = 'native'`).
- **Metrics Engine** — agent metrics ingestion (CPU, memory, disk, network, inventory, heartbeats).
- **Service Checks** — per-service check/retry intervals; standard 0/1/2/3 = OK/Warning/Critical/Unknown.

- **Alert Engine** — rule evaluation and event generation via scheduled `observe:evaluate-alerts`, with channels and monitoring profiles.
- **Capacity Planning** — trend/forecast over collected metrics.
- **Analytics** — performance analytics and reporting.
- **Infrastructure Map** — live topology of hosts/services.

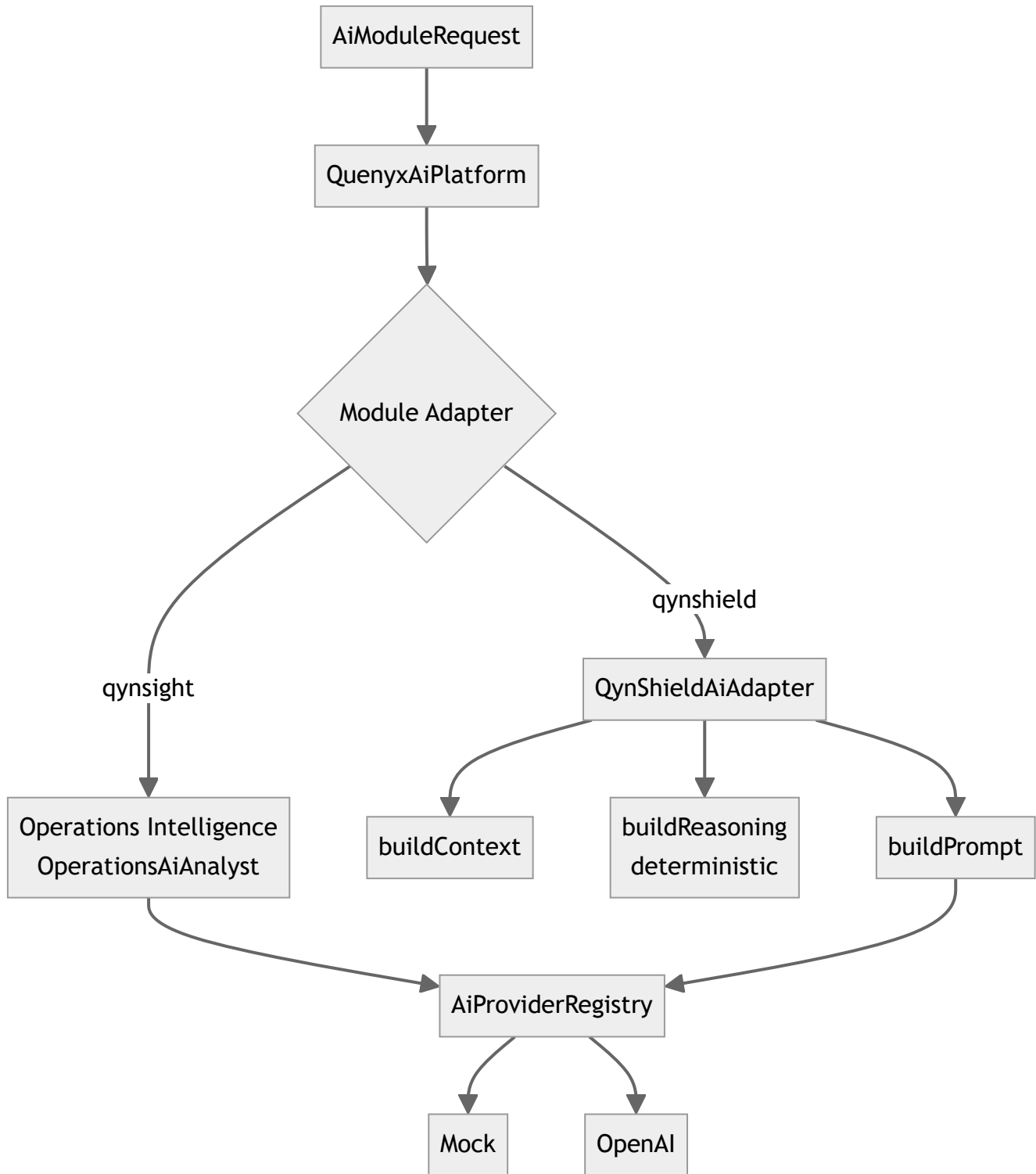
Agents authenticate with enrollment tokens / secrets and push **metrics**, **inventory**, and **heartbeats**; the backend exposes summaries, performance, capacity, alerts (rules/history/channels/profiles), service checks, instances, infra topology, and port scans. Tables are the `observe_*` and `agents*` groups (see Doc 09). Nagios may appear **only** as a legacy migration source or an optional external integration (via Integrations) — never as a platform dependency.

8. QynShield architecture

QynShield is the QCIF compliance engine (see Doc 06 for depth). Layered services: corpus → graph → mapping → evidence → gap → recommendation → copilot → executive, all read-only and **deterministic**, all UUID/provenance-based, all behind `project.qynshield`.

9. Quenyx AI Platform architecture

A shared runtime (see Doc 07) that any module consumes through an **adapter**.



10. API layer

REST/JSON under `/api`. Conventions: sanctum auth, policy + entitlement middleware, per-group throttles (`compliance-copilot`, `compliance-rag`, `compliance-executive`, `ai chat/skills`). Path params use **codes** (framework/release/control/requirement) and **UUIDs** for entities. See Doc 08.

11. Service layer

All business logic is in `app/Services/**` (e.g. `Services/Compliance/**`, `Services/Ai/**`, `Services/QuenyxAI/**`). Controllers validate + delegate; services are deterministic and testable; AI access is funneled through the provider registry only.

12. Database layer

MySQL via Eloquent; **65 migrations** across groups: identity/projects/plans/subscriptions/modules, integrations, audit, QynSight (`observe_*`, agents), jobs, and QCIF (corpus, evidence, gap, recommendation, RAG vectors, AI orchestration, mapping). UUIDs for domain entities; immutability and provenance enforced in corpus tables (see Doc 09).

13. Cache

Laravel cache is used for resolver/scope caching and rate limiting (`RateLimiter` named limiters in `RouteServiceProvider`). Cache driver is environment-configured.

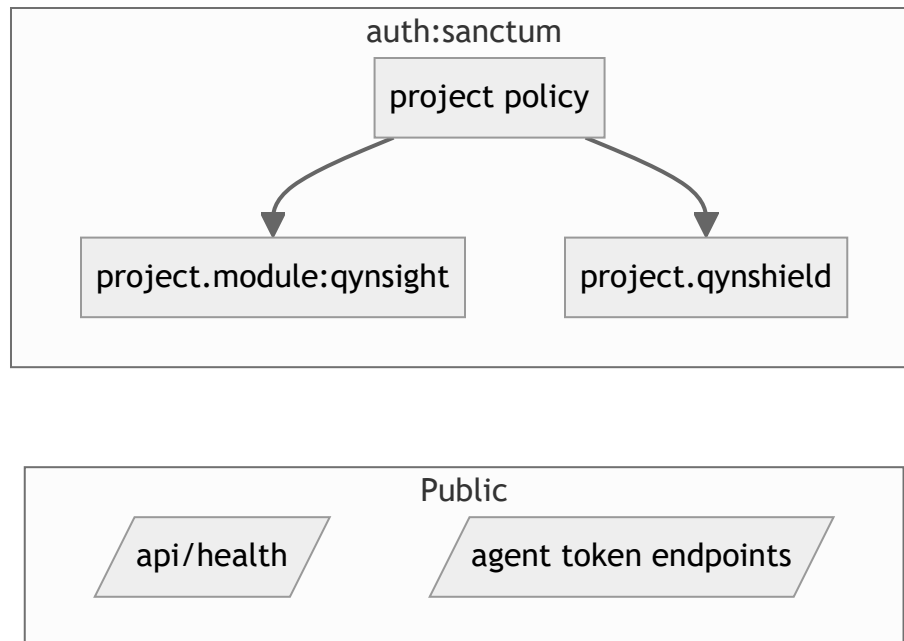
14. Queues

`jobs` and `failed_jobs` tables exist. RAG indexing/deletion are **queue jobs** (`IndexCorpusRevisionForRag`, `IndexRetrievalChunk`, `DeleteVectorIndexForRevision`). A queue worker is required when RAG indexing is enabled (see Doc 10).

15. Audit logging

`audit_logs` table + `AuditLogController`. Sensitive actions (module overrides, executive reads, copilot, etc.) write structured audit entries. Prompt content is **not** logged unless prompt logging is explicitly enabled.

16. Security boundaries



Boundaries: public health; token-authed agent ingestion; everything else sanctum + project policy + module/QynShield entitlement. AI is an additional opt-in layer behind feature flags.

17. Deployment architecture

Ubuntu host(s): Nginx terminates TLS and serves the React build + reverse-proxies `/api` (optionally via the Node gateway) to PHP-FPM (Laravel). MySQL, a cache store, a queue worker, and the Laravel scheduler run alongside. See Doc 10.

18. Current limitations

- QynShield has rich backend/API but an **earlier-stage UI** (executive/demo layer is the current surface).
- Real-model AI and RAG are **feature-flagged**, not GA.
- Audit-sandbox could not run DB/tests/frontend/gateway builds (tooling gaps) — these run on CI/CloudQuenyx (see QA report).
- One low-risk **shadowed duplicate route** (`/ai/chat`) noted in the QA report.

Sprint 20 — Unified AI Workspace (platform layer) — branded "Quenyx AI" (RC1.1)

The **Unified AI Workspace** is a platform-level capability (NOT a business module): a top-level sidebar item beside Dashboard / Workspaces / Integrations. It does not touch `platformRegistry.ts` and the QynSight-only sidebar flag is unchanged.

RC1.1: the surface is presented as **Quenyx AI** (enterprise AI control center). Routes are unchanged for backward compatibility (`/api/ai/*` , SPA `/ai-workspace/*`); a branded `/quenyx-ai/*` alias in `App.tsx` redirects to the canonical routes. The provider list is catalog-driven (`App\Services\Ai\AiProviderCatalog`): a declarative catalog of 14 providers where only those with a real adapter are `executable` (today OpenAI) and the dev-only `mock` provider is hidden outside local/testing. `AiProviderRegistry::defaultKey()` never selects `mock` as the production default.

- **Backend:** flat `/api/ai/*` routes (`routes/ai-workspace.php`), Sanctum + `throttle:ai-workspace` , scoped by a required `workspace` UUID via `AiWorkspaceContextResolver` . Controllers live in `App\Http\Controllers\Ai\Workspace` ; services in `App\Services\Ai\Workspace` . The runtime is **reused** (provider registry, prompt orchestrator, `AiConversationRepository` , skills/capability catalog) — no QynShield logic is duplicated.
- **UUID-only:** `projects` gains an additive, backward-compatible `uuid` (numeric `id` and existing `{project}` bindings unchanged); all AI resources expose UUIDs; audit-derived feed items use deterministic UUIDv5.
- **New tables:** `ai_prompt_templates` , `ai_provider_settings` (encrypted secrets), and `ai_workspace_permissions` (additive per-role overrides over `ProjectPolicy`). `ai_conversations` / `ai_conversation_messages` are reused.
- **RBAC:** `ProjectPolicy::accessAi` / `administerAi` + a fine-grained capability matrix.
- **Frontend:** `AiWorkspaceLayout` + 15 lazy pages under `pages/ai/*` , `aiWorkspaceService` , `useAiWorkspace` hooks, full EN/AR i18n, real empty states; consumes backend APIs only.

Sprint 21 — QynSight Operations Intelligence (additive intelligence layer)

Operations Intelligence makes QynSight a **live AI consumer** of the shared runtime, turning monitoring data into explainable operational intelligence. It is **additive** over the frozen monitoring engine (\$7) and introduces **no new AI services** — all narration goes through the Sprint 20 runtime.

- **Reuse, not duplication.** A single integration point, `App\Services\Observe\Intelligence\OperationsAiAnalyst` , calls the existing `AiProviderRegistry` , `CompliancePromptOrchestrator` , the `AiConversation` / `AiConversationRepository` surface, and `AiAccessAuditLogger` . No new provider registry, prompt/reasoning/RAG engine was created.
- **Deterministic evidence services** (`OperationsEvidenceCollector` , `RootCauseService` , `IncidentTimelineService` , `CapacityAdvisorService` , `InfrastructureImpactService` , `PerformanceAdvisorService` , `AlertExplanationService` , `OperationsIntelligenceService`) read **real** QynSight data and produce evidence; the AI only renders it. Insufficient evidence is stated, never fabricated.
- **UUID-only.** `OperationsEntityId` derives a deterministic UUIDv5 (type + workspace + numeric id) and `OperationsEntityResolver` resolves it back within a workspace — **no schema change**, no

numeric ids exposed.

- **API.** Flat, workspace-scoped routes in `routes/qynsight-intelligence.php` under `/api/qynsight/intelligence/*` (controllers in `App\Http\Controllers\Observe\Intelligence`), Sanctum + `throttle:ai-workspace`, gated by the `qynsight` entitlement, monitoring RBAC, and the `can_use_ai` capability; audited and provider-logged. See Doc 08 §18.
- **Frontend.** `pages/observe/OperationsIntelligence.tsx` dashboard (real data only), reusable `OperationsCopilotDrawer` / `OperationsAlertDrawer` / `QuenyxAiButton` (contextual ✨ actions on host, service, alert, capacity, and infrastructure-map pages), `operationsIntelligenceService`, and full EN/AR i18n. Copilot threads are real Quenyx AI conversations.

AI Adapter Platform + QynAsset Asset Intelligence (Sprint 22)

Sprint 22 generalised module AI into a **reusable adapter framework** and added QynAsset as the **second production AI adapter** — with no per-module branching anywhere.

- **Adapter framework.** `App\Contracts\QuenyxAI\AiModuleAdapter` (coarse capability/action contract)
 - `AbstractAiModuleAdapter` (backward-compatible metadata defaults) + `App\Services\QuenyxAI\AiModuleAdapterRegistry` (singleton discovery/registration/resolution). Modules register one line in `AppServiceProvider::boot()`; the platform discovers them dynamically.
- **Single provider-calling point.** `App\Services\Ai\ModuleAiNarrator` is the only place module intelligence talks to a provider (reusing `AiProviderRegistry`, `CompliancePromptOrchestrator`, `AiAccessAuditLogger`). Sprint 21's `OperationsAiAnalyst` now **delegates** to it (behavior preserved). No duplicated provider/prompt/reasoning/RAG/orchestration.
- **Discovery API.** `/api/ai/adapters`, `/api/ai/adapters/{module}`, `/api/ai/adapters/capabilities`, `/api/ai/actions` — entitlement-filtered, RBAC-gated, audited (Doc 08 §19).
- **QynAsset.** An asset is a **discovered host** (`observe_targets_hosts` + `agents` + `agent_inventories`); hardware reuses Capacity Planning, dependencies reuse the Infrastructure Map. Services in `App\Services\Asset\Intelligence\*`; UUID-only via `AssetEntityId`; API in `routes/qynasset-intelligence.php`. License/lifecycle-date facts have no source and are reported as **not collected** — never fabricated. See Docs 22 and 23.

Sprint 23 — Automation Platform & Incident Workspace

Automation Platform (shared). A registry-driven automation layer under `App\Services\Automation\*`, designed so every future module consumes it instead of duplicating execution logic. Core pieces:

- **AutomationAdapterRegistry** + **ExecutionAdapter** contract — dynamic registration of runners (Script, REST, Webhook, SSH, PowerShell shipped; Docker/Kubernetes/OCI/AWS/Azure/GCP register as `PlannedExecutionAdapter`s). No hardcoded execution; the UI Library reflects the registry.

- **ActionRegistry** — catalog of reusable actions (schema + `destructive / rollback` flags).
- **ExecutionEngine** — dry-run by default, approval gate for live, timeout/retry, per-step recording, audit, and learning. DTOs: `ExecutionContext` , `ExecutionResult` (immutable).
- **WorkflowEngine / RunbookEngine** — validate data-only definitions and dispatch via the Execution Engine. `ApprovalEngine` , `RollbackEngine` , `ExecutionHistory` , `AutomationAuditLogger` , `AutomationLearningService` complete the platform.

Safety envelope. Global `automation.live_execution` master switch (default OFF), HTTP host allowlist, per-runner enable flags, mandatory human approval for live, adapter `rollback` , workspace isolation, RBAC (`accessAi read / administerAi privileged`), UUID-only, full audit. Adapters never fabricate — they return `dry_run` or honestly `skipped` .

QynRun is the operator surface (`App\Http\Controllers\Automation\*` , `routes/qynrun-automation.php`); **QynReact** is the **Incident Workspace** (`App\Services\Incident\*` , `routes/qynreact-incident.php`) that aggregates Timeline, cross-module context (`CrossModuleOrchestrator` over the AI Adapter Registry — never module-branch), Automation, Knowledge, Evidence, Resolution, and Postmortem. New tables in `2026_06_30_010000_create_automation_and_incident_tables.php` . See Docs 24–27.

Sprint 24 — Enterprise Knowledge & Collaboration Platform

Knowledge as a shared platform capability. QynKnow, QynSupport, and QynNotify are unified so that Knowledge, Search, Timeline, and Collaboration are platform services every module reuses — not isolated per-module implementations. Core pieces under `App\Services\Knowledge\*` , `App\Services\Collaboration\*` , `App\Services\Support\*` , `App\Services\Notification\*` :

- **KnowledgeSourceRegistry** + **KnowledgeSource** contract — dynamic, registry-driven sources. `InternalKnowledgeBaseSource` is live (operates on `knowledge_documents` with deterministic lexical search); Markdown/PDF/HTML/Git/Confluence/SharePoint/Google Drive/OneDrive/GitHub & GitLab Wiki/ MediaWiki/Elastic-OpenSearch/Vector Store register as `PlannedKnowledgeSource s` (`isOperational()` $\equiv$ `false`). **No provider-specific branching** — the registry is the only seam.
- **EnterpriseSearchService** — one keyword + semantic search interface over **real indexed rows** (knowledge documents, incidents, tickets, notifications, automation workflows/runbooks); unified, deterministically scored results. No fabricated hits.
- **KnowledgeGraphService (v2)** — deterministic, bounded read-model of typed nodes (project, hosts, assets, alerts, incidents, runbooks, automation, tickets, documents, notifications, people, ...) and edges, with traversal.
- **GlobalTimelineService** — platform-wide chronological read-model aggregating incidents, automation, tickets, notifications, and knowledge updates from real tables (writes nothing).
- **CollaborationService** — polymorphic comments/mentions/participants (`watcher / assignee / owner`) addressed by (`entity_type` , `entity_uuid`) for **any** entity. One shared layer; no per-module copy.
- **Ticket Intelligence** (`TicketIntelligenceService`) and **Notification Intelligence** (`NotificationIntelligenceService` + deterministic `NotificationService` ingest: dedup,

correlation, urgency scoring, recipient/channel selection, escalation) — evidence-based, honest about "insufficient evidence", real recipients only (no fake routing).

AI envelope. Every new intelligence surface narrates exclusively through the shared `ModuleAiNarrator` (no direct provider calls), discovered via the **AI Adapter Registry** (`QynKnowAiAdapter` , `QynSupportAiAdapter` , `QynNotifyAiAdapter`); cross-module context reuses `CrossModuleOrchestrator` (never module-branch). All drafts are editable and never auto-applied. Workspace isolation, RBAC (`accessAi` / `can_use_ai`), UUID-only, and full audit (`PlatformAuditLogger`) throughout. New tables in `2026_07_05_010000_create_knowledge_collaboration_tables.php` . See Docs 28–32.